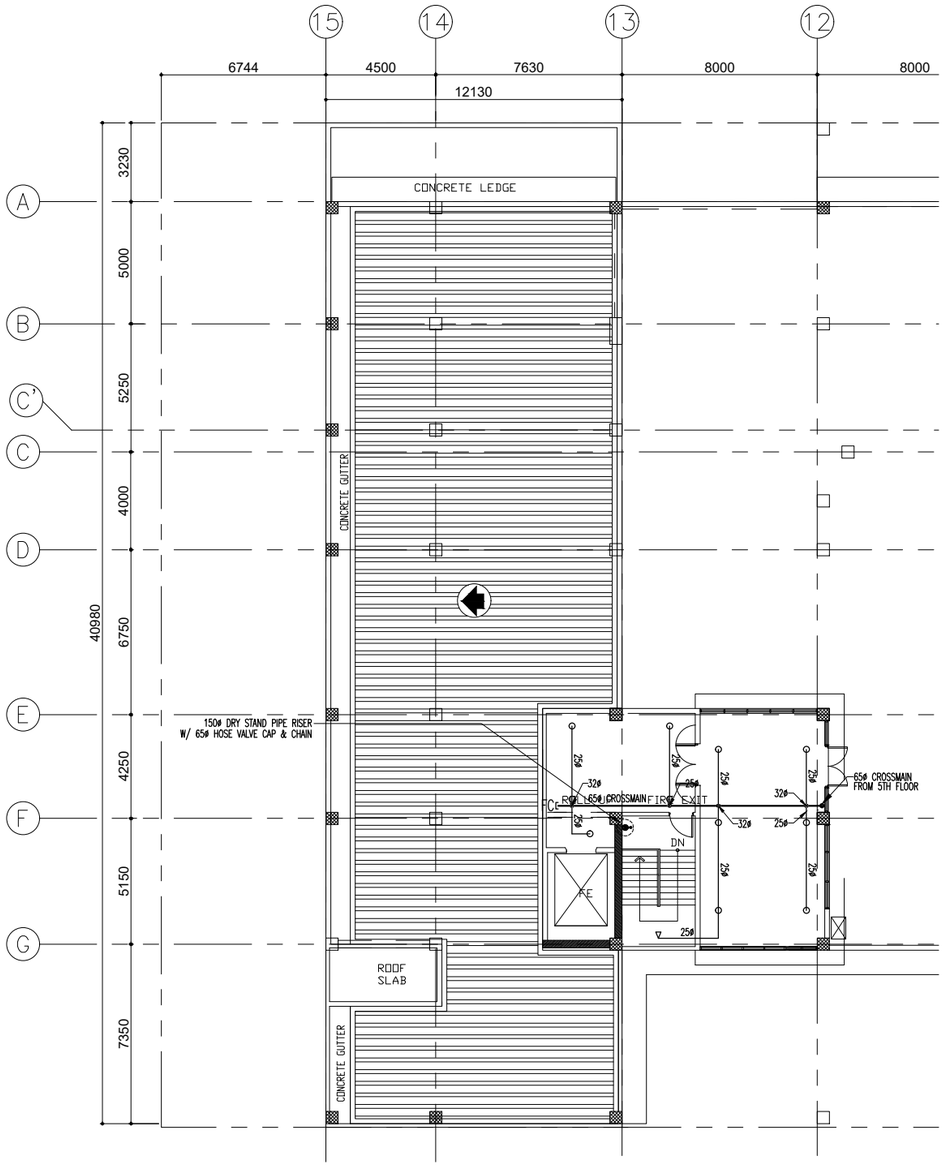
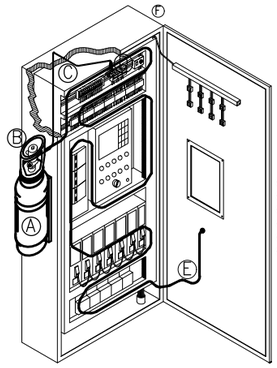


SIXTH/RD FLOOR PLAN
FIRE PROTECTION LAYOUT
A FP-07 SCALE 1:100



A NOVEC 1230 EXTINGUISHING AGENT CYLINDER
B PRESSURE DIFFERENTIAL VALVE
C DISCHARGE PIPE
D DISCHARGE NOZZLE
E SENSOR TUBING
E CONTROL/ELECTRICAL PANEL

SEQUENCE OF OPERATION :

1. THE FLEXIBLE SENSOR TUBING IS INSTALLED DIRECTLY ABOVE THE EQUIPMENTS, WIRES AND CIRCUITRY WHERE A FIRE COULD START. WHEN IN SERVICE, THE TUBING IS PRESSURIZED WITH NOVEC 1230 TO 16BAR. THE DYNAMICS OF PRESSURIZATION MAKE THE TUBING MORE REACTIVE TO HEAT.
2. WHEN FLAME-UP OCCURS, THE HEAT OF THE FIRE CAUSES THE PRESSURIZED SENSOR TUBE TO BURST AT THE HOTTEST SPOT (APPROXIMATELY 110°C).
3. THE SUDDEN TUBE DEPRESSURIZATION ACTUATES THE SPECIAL PRESSURE DIFFERENTIAL VALVE AND INSTANTLY FLOODS THE ENTIRE ELECTRICAL CABINET AREA WITH NOVEC 1230 EXTINGUISHING AGENT. THE FIRE IS QUICKLY SUPPRESSED JUST MOMENTS AFTER FIRE BEGAN.

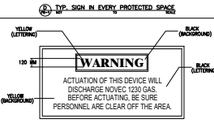
H FP-07 CLEAN AGENT (NOVEC 1230) ELECTRICAL CABINET FIRE SUPPRESSION SYSTEM
SCALE: NTS

WS-1 - WARNING SIGN 1

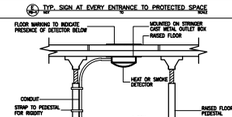
WS-2 - WARNING SIGN 2

G FP-07 TYP. SIGN IN EVERY PROTECTED SPACE
SCALE: NTS

G FP-07 TYP. SIGN AT EVERY ENTRANCE TO PROTECTED SPACE
SCALE: NTS

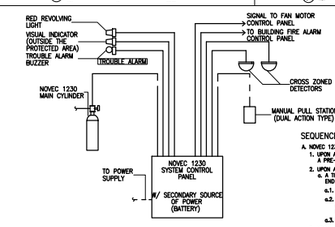


WS-3 - WARNING SIGN 3



G FP-07 TYP. SIGN IN MANUAL PULL STATION
SCALE: NTS

G FP-07 HEAT/SMOKE DETECTOR MOUNTING DETAIL @ UNDER FLOOR IF APPLICABLE
SCALE: NTS



NOTES :
1. ROOM SHALL BE SEALED TIGHT TO PREVENT GAS LEAKAGE UPON DISCHARGE OF THE SYSTEM.
2. PIPE BETWEEN CYLINDER (CABINET) TO THE CYLINDRICAL TANK MUST INCLUDE THE DISCHARGE OPERATION OF THE SYSTEM.
3. FOR ANCHOR TANGENTIAL SUPPORTS SHALL SUBMIT SHOP DRAWINGS & CALCULATION TO DETERMINE FIXED PIPE & NOZZLE SIZES.

SEQUENCE OF OPERATION :

A. NOVEC 1230 CONTROL PANEL
1. UPON ACTUATION OF FIRST ZONE DETECTORS (PRE ALARM MODE), A PRE-DISCHARGE ALARM SHALL SOUND.
2. UPON ACTUATION OF THE SECOND DETECTOR :
a. A TIME DELAY (30 SECONDS) SHALL START AT THE END OF WHICH,
a.1. POWER IN THE AFFECTED AREA WILL SHUTDOWN.
a.2. A VISUAL INDICATOR OUTSIDE THE ENTRANCE TO THE AREA OR ZONE MUST ACTIVATE INDICATING THAT RE-ENTRY IS PROHIBITED.
a.3. DISCHARGE OF THE GAS SHALL START.
3. UPON ACTIVATION OF MANUAL STATION, ALL FUNCTION UNDER 1 & 2 SHALL OCCUR INCLUDING TIME DELAYED DISCHARGE.

1 FP-07 NOVEC 1230 CONTROL PANEL SCHEMATIC DIAGRAM
SCALE: NTS

GRAPHIC SCALE:
0 2 4 6 8
SCALE 1:100 M
NOTE: CHECK GRAPHIC SCALE PRIOR TO SCALING

CONSULTANT		OWNER	APPROVED BY:	PROJECT TITLE	SHEET CONTENT/S :	ISSUE NO.	REVISIONS	NO.	DATE		DATE	SHEET NO.
E. H. SISON ENGINEERS, CO. ENGINEERS * PLANNERS * CONSULTANTS 3431 GUERNICA ST., PALANAN, MAKATI CITY TEL. Nos. 8317115, 8337030	RAMON D. AGUILOS PROFESSIONAL MECHANICAL ENGINEER PTR NO.: 2464195 D — PRC NO. : 2736 DATE : 01-05-22 — TIN. NO. : 137-133-799 PLACE : QUEZON CITY	PDIC	CYRUS T. GALANG VICE PRESIDENT ADMINISTRATIVE SERVICES GROUP	RENOVATION OF PDIC BUILDING IN CHINO ROCES LOCATION: 2228 CHINO ROCES AVENUE, MAKATI CITY	ROOF DECK FLOOR PLAN FIRE PROTECTION LAYOUT FIRE SUPPRESSION SCHEMATIC DIAGRAM AND WARNING SIGNS					DESIGNED BY: RDAEC CADD BY: RJA CHECKED BY: APPROVED BY: SCALE: AS SHOWN		FP-07